1 Diagonalizable Operators

We saw last time that a linear operator $T \in \mathscr{L}(V)$ on a finite-dimensional vector space V will have a diagonal matrix $[T]_B$ with respect to a basis B if and only if each vector in B is an eigenvector of T.

Definition 1.1. A linear operator $T \in \mathscr{L}(V)$ is said to be *diagonalizable* if there exists a basis consisting of eigenvectors of T. Such a basis is called an *eigenbasis*.

We will now prove some results which will help us determine when such a basis exists.

Proposition 1.2. *If $T \in \mathscr{L}(V)$, then every set of eigenvectors corresponding to distinct eigenvalues of T is linearly independent.*

Proof. Suppose the claim is actually false. Then there exists a smallest positive integer m such that there exists a linearly dependent set $\{v_1, \ldots, v_m\}$ of eigenvectors of T corresponding to distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_m$ (note that $m \geq 2$ because an eigenvector is, by definition, nonzero). Thus there exist scalars $c_1, c_2, \ldots, c_m$, none of which are zero (because of the minimality of m) such that

$$c_1 v_1 + \cdots + c_m v_m = 0.$$

Exercise 1.3. Apply $T - \lambda_m \mathrm{id}_V$ to both sides of the equation and use the result to complete the proof.

□

Corollary 1.4. *If $\dim V = n$ and $T \in \mathscr{L}(V)$ has n distinct eigenvalues, then T is diagonalizable.*

Exercise 1.5. Prove Corollary 1.4.

Exercise 1.6. Is $T \in \mathscr{L}(F^3)$, defined by $T(x,y,z) = (2x+y-z, 3y+z, -z)$, diagonalizable?

We now consider the case where an eigenvalue occurs with multiplicity greater than one. We first distinguish two different notions of multiplicity.

Definition 1.7.

(a) The multiplicity of λ_k as a root of the characteristic polynomial of A is called the *algebraic multiplicity* of λ_k.

(b) The dimension of the eigenspace corresponding to λ_k is called the *geometric multiplicity* of λ_k.

Lemma 1.8. *For each $1 \leq k \leq m$, the geometric multiplicity of λ_k is less than or equal to the algebraic multiplicity of λ_k.*

Proof. Suppose $T \in \mathscr{L}(V)$ has an eigenvalue λ_k. Let $d_k = \dim E_{\lambda_k}$. Choose d_k linearly independent eigenvectors with eigenvalue λ_k. If we choose a basis for V where the first d_k vectors are these eigenvectors, then the matrix of T with respect to this basis will take the form

$$[T] = \begin{bmatrix} \lambda_k I_k & B \\ 0 & C \end{bmatrix}$$

where I_k is the $d_k \times d_k$ identity matrix. The characteristic polynomial of T can therefore be factored as

$$(\lambda - \lambda_k)^{d_k} \det(\lambda I - C) \tag{1.9}$$

so the algebraic multiplicity of λ_k is at least d_k. □

Exercise 1.10. As a concrete example of the argument above, show that the characteristic polynomial of the matrix

$$A = \begin{pmatrix} \lambda_1 & 0 & 0 & 0 & 0 & a & b \\ 0 & \lambda_1 & 0 & 0 & 0 & c & d \\ 0 & 0 & \lambda_2 & 0 & 0 & e & f \\ 0 & 0 & 0 & \lambda_2 & 0 & g & h \\ 0 & 0 & 0 & 0 & \lambda_2 & i & j \\ 0 & 0 & 0 & 0 & 0 & k & l \\ 0 & 0 & 0 & 0 & 0 & m & n \end{pmatrix}$$

takes the form in Equation (1.9).

Exercise 1.11. Find the geometric and algebraic multiplicities of the eigenvalues of $T \in \mathscr{L}(F^3)$, defined by $T(x,y,z) = (3x+y-z, 2x+2y-z, 2x+2y)$.

Proposition 1.12. *Suppose $T \in \mathscr{L}(V)$ and $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T. Then*

$$E_{\lambda_1} + \cdots + E_{\lambda_m}$$

is a direct sum. Furthermore, if V is finite-dimensional, then

$$\dim E_{\lambda_1} + \cdots + \dim E_{\lambda_m} \leq \dim V.$$

Exercise 1.13. (a) Explain why the first claim in Proposition 1.12 follows from Proposition 1.2. (*Hint*: Suppose $v_1 + \cdots + v_m = 0$, where $v_k \in E_{\lambda_k}$ for each $k = 1, \ldots, m$.)

(b) Prove the second claim in Proposition 1.12.

The next theorem gives several conditions which are equivalent to diagonalizability of $T \in \mathscr{L}(V)$.

Theorem 1.14 (Conditions equivalent to diagonalizability)**.** *Suppose V is finite-dimensional and $T \in \mathscr{L}(V)$. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of T. Then the following are equivalent.*

(a) V has a basis consisting of eigenvectors of T

(b) $V = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_m}$

(c) $\dim V = \dim E_{\lambda_1} + \cdots + \dim E_{\lambda_m}$

(d) The geometric multiplicity of each eigenvector is equal its algebraic multiplicity.

Exercise 1.15. Prove that (a) $\implies$ (b) $\implies$ (c) in Theorem 1.14.

Exercise 1.16. Prove that (c) $\iff$ (d) in Theorem 1.14.

We now show that (c) $\implies$ (a) in Theorem 1.14 to complete the proof.

Proof. Suppose (c) holds. Choose a basis of each E_{λ_k}. Put all of these together to form a set $\{v_1, \ldots, v_n\}$ of eigenvectors of T, where $n = \dim V$. To show that this set is linearly independent, suppose

$$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0,$$

where $\alpha_1, \ldots, \alpha_n \in F$. For each $k = 1, \ldots, m$, let u_k denote the sum of all the terms $\alpha_j v_j$ such that $v_j \in E_{\lambda_k}$. Thus each u_k is in E_{λ_k} and

$$u_1 + \cdots + u_m = 0.$$

Since $u_1 + \cdots + u_m \in E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_m}$, this implies that $u_1 = \cdots = u_m = 0$. It follows that for each k, the sum of the terms $\alpha_j v_j$ with v_j a basis of E_{λ_k}. Since the v_j are linearly independent, this implies that $\alpha_j = 0$ for all j. This shows that $\{v_1, \ldots, v_n\}$ is linearly independent and hence a basis of V. Hence, (c) $\implies$ (a). $\square$

While every operator on a nonzero finite-dimensional complex vector space has an eigenvector, a given operator may not have enough eigenvectors to be diagonalizable, as is shown in the next example.

Example 1.17. Define $T \in \mathscr{L}(V)$ by $T(x,y,z) = (y,z,0)$. The matrix of T with respect to the standard basis is

$$[T] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The characteristic equation is then

$$\begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 0 & 0 & -\lambda \end{vmatrix} = -\lambda^3 = 0$$

which implies that $\lambda = 0$ is the only eigenvector. The corresponding eigenvector $v = (v_1, v_2, v_3)$ is a solution of the system

$$\begin{aligned} v_2 &= 0 \\ v_3 &= 0, \end{aligned}$$

hence $E_0 = \text{span}\{(1,0,0)\}$. Since $\dim E_0 = 1 < 3$, it follows from Theorem 1.14 that T is not diagonalizable.

Exercise 1.18. Is $T \in \mathscr{L}(F^3)$, defined by $T(x,y,z) = (3x+y-z, 2x+2y-z, 2x+2y)$ diagonalizable?

Example 1.19. Let

$$A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}.$$

The characteristic equation of A is

$$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = (\lambda - 1)(\lambda - 2)^2 = 0$$

and therefore the distinct eigenvalues of A are $\lambda = 1, 2$. We find

$$E_1 = \text{span}\left\{ \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\},\ W_2 = \text{span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Since $\dim E_1 + \dim E_2 = 1 + 2 = 3 = \dim V$, by Theorem 1.14 A is diagonalizable. Choosing the basis

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

we have

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Exercise 1.20. Let

$$A = \begin{bmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{bmatrix}.$$

Diagonalize A, if possible.